

MMC Watershed Rainfall Report

Rainfall observations for the Moores Mill Creek watershed

MMC Watershed Data

August 2, 2026

MMC Watershed Data collects and standardizes rainfall observations from Auburn and Opelika stations near the Moores Mill Creek watershed. This report compares station totals and 10-minute rainfall patterns for one saved collection selected at render time.

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1 Purpose and reporting period

MMC Watershed Data retrieves rainfall observations from the City of Auburn LI-COR network and the City of Opelika THOR network, preserves the API responses as evidence, validates their structure, and converts them into comparable station CSVs. This report uses the **latest saved MMC collection**, with the reporting period read from its saved processed

CSV filenames. The selected period is **May 01, 2026 through June 01, 2026**, with **38,285 validated observations** from **9 stations**. All configured stations are represented. It asks: **which station recorded the greatest total interval rainfall during this period, and how did rainfall vary among the represented stations?**

2 Station and watershed location

The Moores Mill Creek (MMC) watershed is the land area where surface runoff drains toward Moores Mill Creek. Its boundary defines the geographic area of interest for this project. Figure 1 places the configured Auburn and Opelika rainfall stations relative to that boundary: blue points are Auburn stations, red points are Opelika stations, and the green polygon is the watershed.

Each station measures rainfall at one location. Stations inside and around the watershed help show how rainfall varies across the area, but an individual station value should not be interpreted as rainfall over the entire watershed.

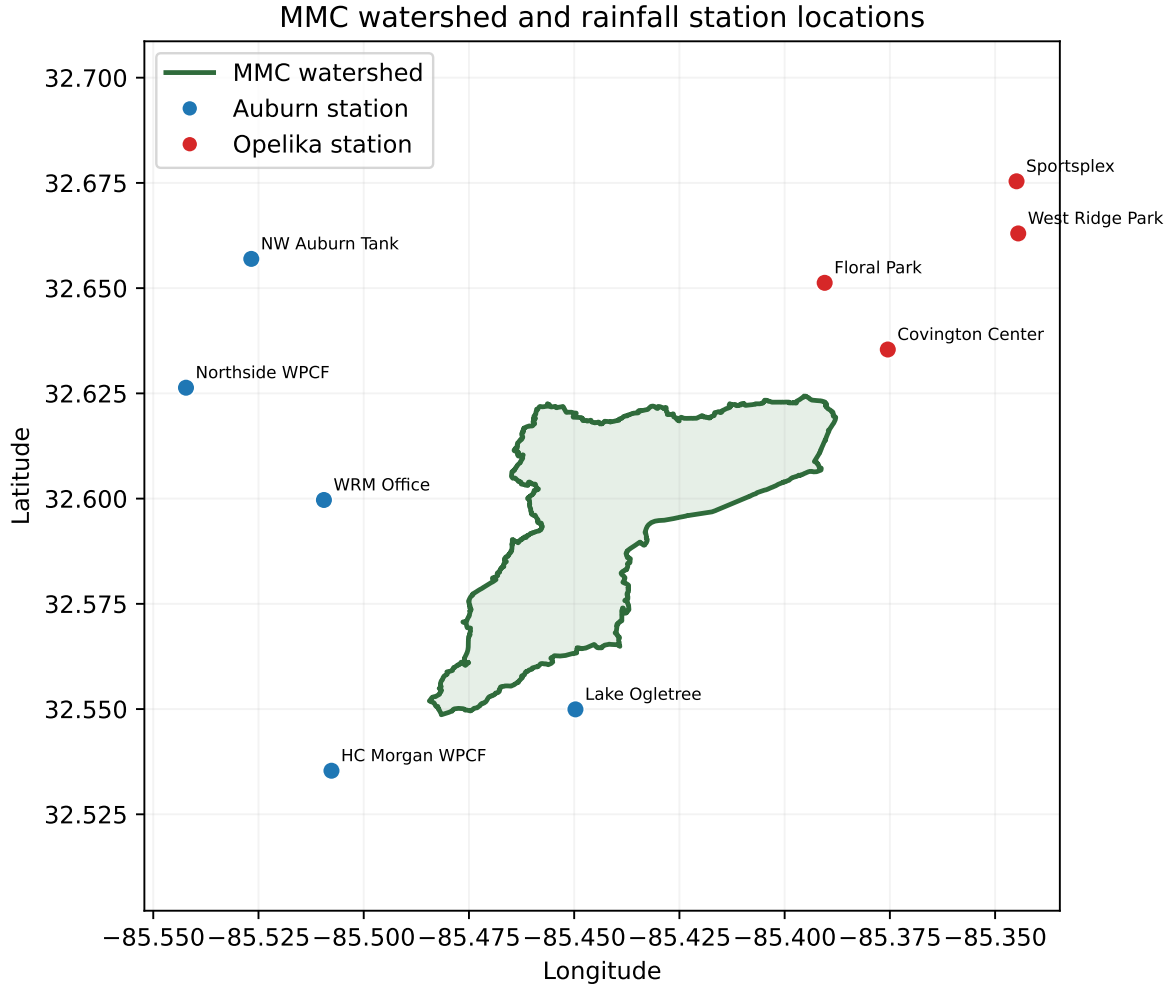


Figure 1: Auburn and Opelika rainfall stations relative to the Moores Mill Creek watershed boundary.

3 Data and method

The report selects one complete processed collection using the date range encoded in MMC's station filenames. It always chooses the collection most recently written under `data/processed/`, so its period follows the latest API collection made with `mmc`. If no generated collection exists, the report uses the bundled dataset under `reports/data/` so the repository PDF remains reproducible offline.

All records are loaded through `mmc_watershed_data.reporting`, which calls the project's typed `load_rainfall_records` function. The report does not recreate Auburn epoch conversion,

Opelika cumulative-counter conversion, Pydantic validation, or API requests. Rainfall totals are sums of processed interval `RainIn` values in inches. The processed CSVs already contain one `RainIn` value per source observation at a nominal 10-minute cadence. The station time-series figures plot those rows directly at their saved `Date_hour` timestamps; the report does not resample or sum them into new time windows.

4 Results

4.1 Summary

For **May 01, 2026 through June 01, 2026**, **Floral Park** recorded the largest station total at **8.92 inches**. That represents **16.3%** of the sum across represented stations. Auburn stations contributed **19.50 inches** and Opelika stations contributed **35.07 inches** when station interval values are summed.

4.2 Station statistics

Table 1: Processed rainfall statistics by station for the selected reporting period.

Station	City	Total rain (in)	Wet rows	Max interval (in)
Floral Park	Opelika	8.92	207	0.67
West Ridge Park	Opelika	8.88	189	0.60
Covington Center	Opelika	8.65	197	0.49
Sportsplex	Opelika	8.62	198	0.49
WRM Office	Auburn	4.44	199	0.28
NW Auburn Tank	Auburn	4.43	211	0.20
HC Morgan	Auburn	4.35	201	0.23
WPCF				
Northside WPCF	Auburn	4.09	198	0.21
Lake Ogletree	Auburn	2.19	108	0.23

4.3 Rainfall totals

Blue bars identify Auburn stations and red bars identify Opelika stations, matching the city colors used by the dashboard. The legend makes that encoding explicit.

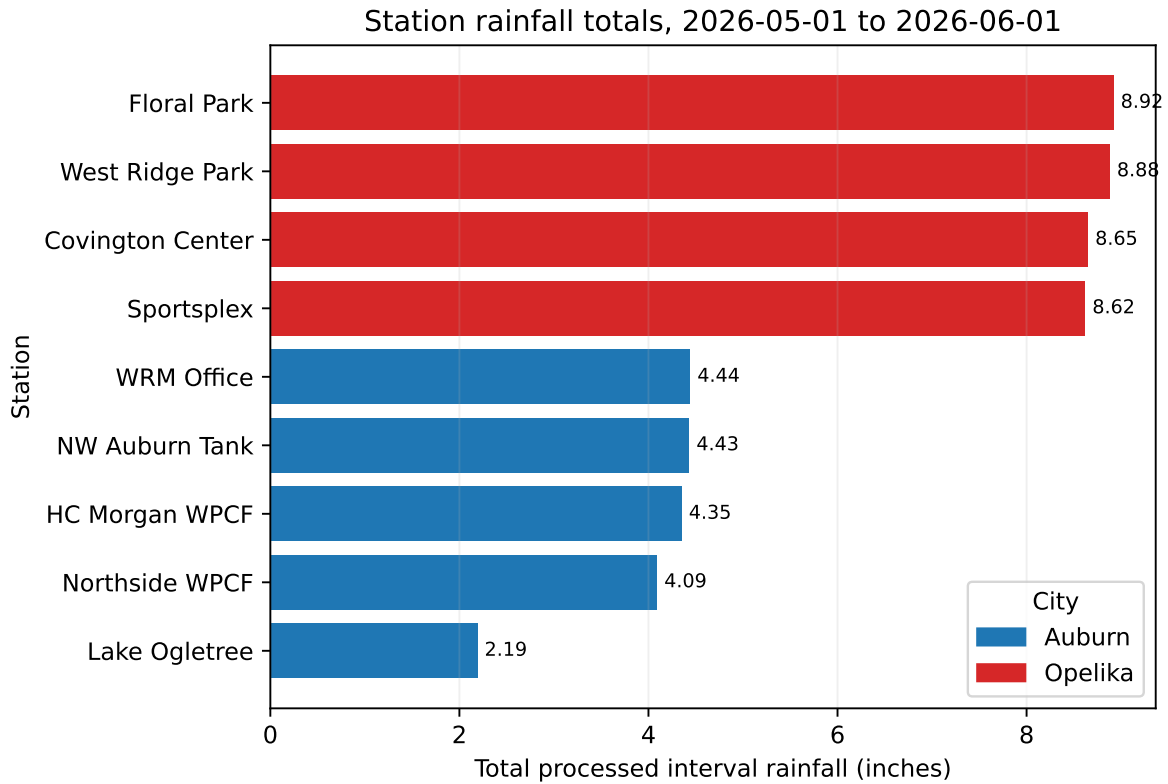


Figure 2: Total processed interval rainfall by station for the selected reporting period.

4.4 Rainfall at 10-minute intervals

Each panel below represents one station over the full selected period. Every bar reports one `RainIn` value directly from the processed CSV; the shared horizontal scale makes the timing of rainfall comparable across stations.

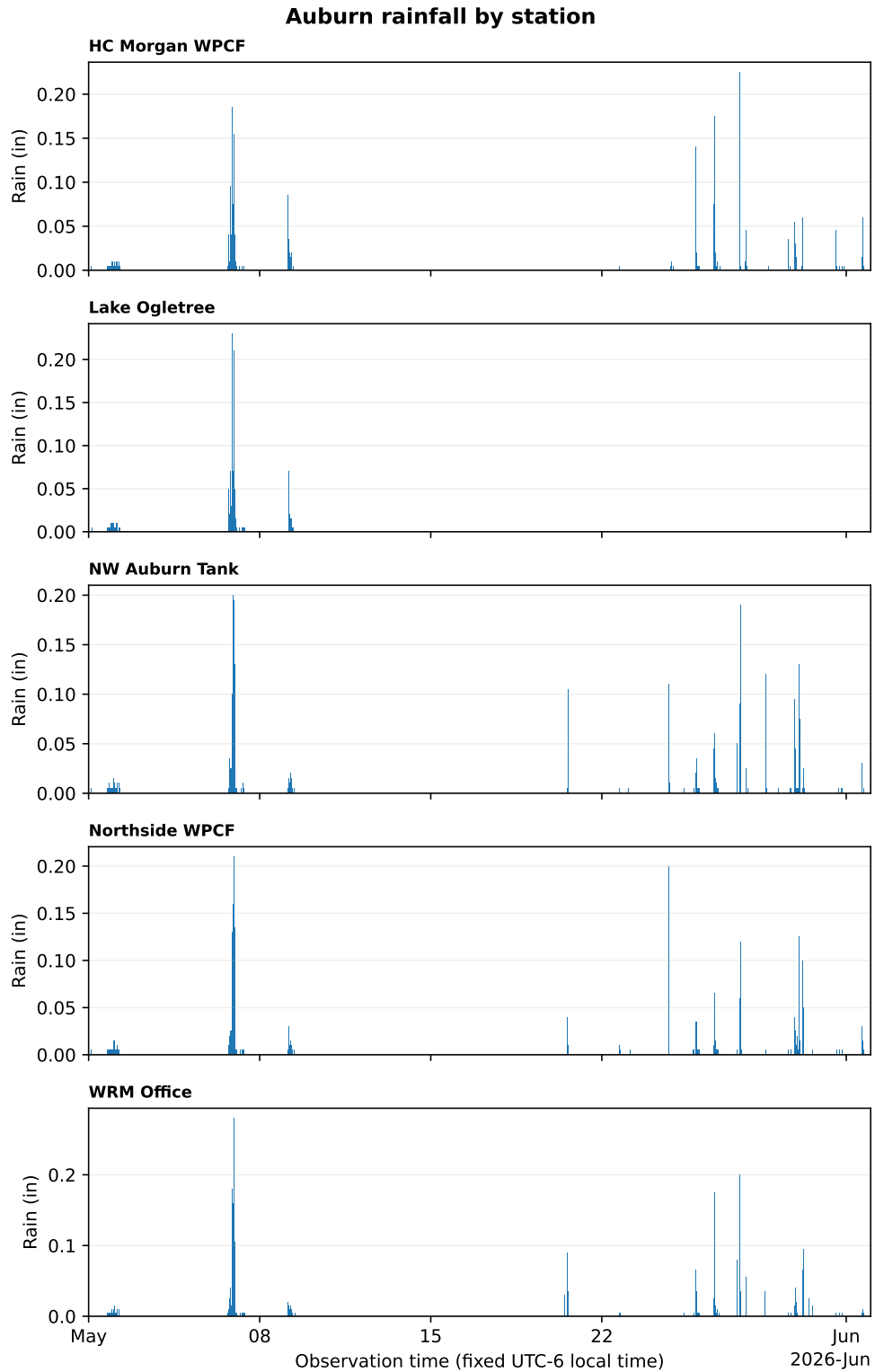


Figure 3: Auburn processed rainfall observations at a nominal 10-minute cadence during the selected reporting period.

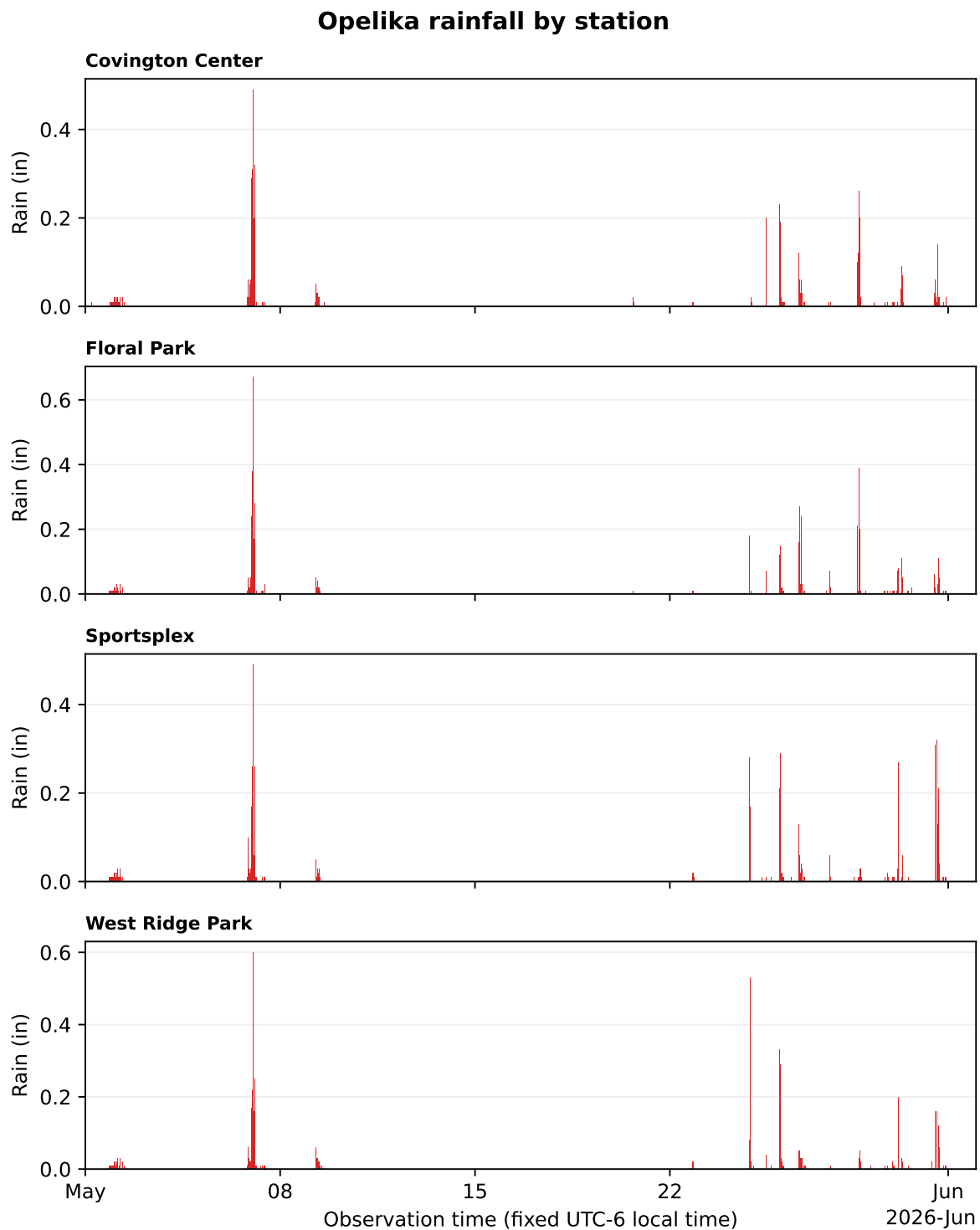


Figure 4: Opelika processed rainfall observations at a nominal 10-minute cadence during the selected reporting period.

4.5 Interpretation and limitations

The selected records show that **Floral Park** had the largest accumulated interval total during **May 01, 2026 through June 01, 2026**. The comparison describes the observations available in the selected collection; it does not estimate rainfall between stations or prove that one part of the watershed received more rainfall everywhere. Auburn and Opelika can have different source sampling behavior, and station failures or empty responses reduce coverage.

5 Provenance and reproducibility

Table 2: Data and software provenance recorded for this report render.

Field	Recorded value
Selected input	latest saved MMC collection
Requested period	2026-05-01 through 2026-06-01
Processed station files	9
Validated observations	38285
Stations represented	9
Rainfall unit	inches per processed interval
Auburn source	https://www.licor.cloud/api/v2/timeseriesdata
Opelika source	https://360.thormobile.net/thorcloud/api/weatherpacketsbyinterval
Project version	0.5.0
Python	3.13.11
Rendered (UTC)	2026-08-02T23:53:55+00:00

After collecting a period, the report automatically selects the most recently written processed collection:

```
uv run mmc --start-date 2026-01-01 --end-date 2026-01-08
quarto render reports/mmc-rainfall-report.qmd --to pdf
```

The report source, inputs, PDF, metadata, and logs contain no API key or other secret.